Which Information Reduces Epistemic Uncertainty? Identification Limits in Source Mixtures

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Abstract

Epistemic uncertainty is reducible through additional information, but which information? REACH studies this question in a finite-cell source-mixture model. We retain source prevalences compatible with unlabelled deployment inputs and report distinct widths for known-source membership, a reference-anchored target probability, and candidate-novel membership. Exact constructions show that perfect input-law knowledge can leave positive membership widths when mixtures are not identifiable. Synthetic acquisition sweeps with estimated known-source conditionals compare unlabelled inputs, source-stratified reference target labels, and reliable deployment source identities. Inputs narrow membership widths under distinguishable sources; target labels reduce the reference-anchored label width; source identities reduce widths below input-only identification floors under confounding. Improving reference inputs can affect several widths, so these are acquisition-dependent responses of distinct targets, not an additive decomposition or an exclusive assignment of acquisitions. The candidate-source distribution and label-noise rate are specified. Label noise is a controlled nuisance factor; aleatoric uncertainty is not estimated. The results characterize identification limits and acquisition contrasts within the declared model.

[Mateus: I have the impression that the abstract is very compressed. Even though the statements express our findings, I suggest to us to maintain a narrative like: problem → gap → method → findings → implications]

Preliminary work. Under review by AISTATS 2027. Do not distribute.

1 INTRODUCTION

"Collect more data" is incomplete advice for reducing epistemic uncertainty. An acquisition can identify the observable input distribution without identifying the source mixture that generated it. Uncertainty that persists under logging may therefore require different information rather than more inputs. Our question is which acquisition reduces uncertainty in a specified target.

REACH retains source mixtures compatible with the available evidence and propagates the set into posterior intervals. We study two known sources and a declared candidate novel source, distinguishing source identity Z from a binary target Y. Reference target labels are collected within already known source strata; deployment source identities instead reveal the incoming mixture composition. Figure 1 illustrates this distinction. All numerical experiments are synthetic. **[Mateus: This statement about the numerical experiments being synthetic goes against our usage of CIFAR, no?] [Mateus: I think we could explicitly say what is our hypothesis, main objective, and sub-objectives, pointing out against our hypothesis.]**

The contribution has two parts. Exact equivalent-world constructions establish positive input-only identification widths for the reported membership targets. Controlled sweeps then show how acquisitions intersect those ambiguities and how reference uncertainty creates cross-effects. Figure 3 is the main empirical comparison; Figure 2 explains its floors. Supporting baseline comparisons and a finite-pool stress test are in the appendix. We use standard confidence bounds and claim no general contraction theorem or new prevalence estimator. **[Mateus: This last paragraph appears also to be a little bit compressed. Maybe**

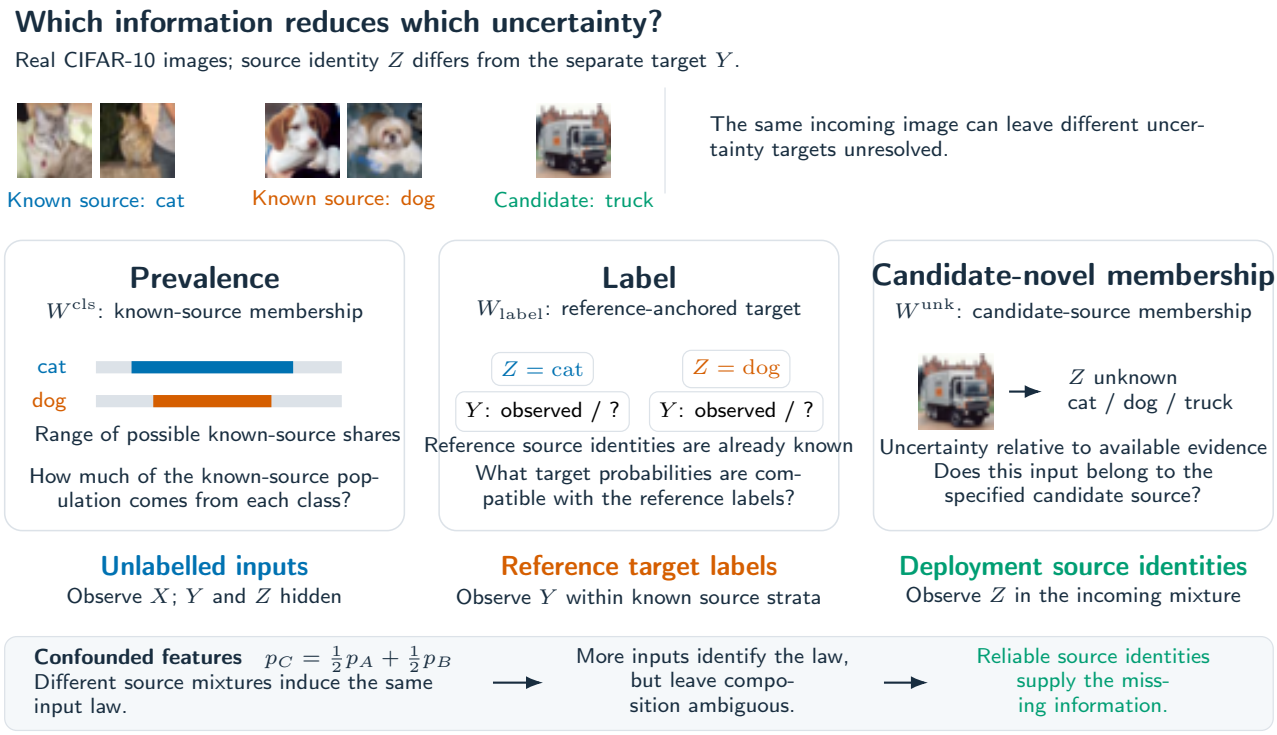


Figure 1: Information and uncertainty targets. CIFAR-10 thumbnails (Krizhevsky, 2009) illustrate known sources (cat/dog) and a specified candidate (truck). Y is a separate hypothetical binary target, not a CIFAR class annotation. Share bars motivate prevalence; W^{cls} is a per-input membership width. The confounding equality is schematic, not a measured CIFAR property. Reference labels use known source strata; deployment identities sample the incoming mixture. Column alignment implies neither exclusive acquisition effects nor additivity.

instead of focusing on the figures, we should point out the manuscript sections]

2 MODEL, EVIDENCE, AND TARGETS

[Mateus: Being very honest, I barely grasp the math part behind the method. But this is my own limitation. I feel also we could explain in more detail how REACH methodology works and its pipeline. Also argue in more detail why we chose to analyze those 3 different categories.] Let C be a fixed input cell, $Z \in \{0, 1, q\}$ the source, and Y a binary target. The row-stochastic matrix A contains source cell laws; $w \in \Delta_3$ gives deployment prevalences, and $\mu = A^\top w$ is the deployment cell law. Known-source target rates are $\theta_k = \Pr(Y = 1 \mid Z = k)$, with $Y \perp C \mid Z$. The candidate row A_q is specified exactly. REACH does not discover arbitrary novel classes.

Input and reference evidence. For n iid deployment cells with empirical law $\hat{\mu}$, define

$$e(N, \beta) = \frac{1 + \sqrt{2\log(1/\beta)}}{\sqrt{N}}. \tag{1}$$

The Euclidean error is at most $e(n, \alpha_{\log})$ with probability at least $1 - \alpha_{\log}$. A bounded-difference argument is given in Section A. With estimated known-source rows $\hat{A}_k$, let $b = \max_{k=0,1} e(r, \alpha_{\mathrm{ref}}/2)$ and use

$$\mathcal{W} = \{w \in \Delta_3 : \|\hat{A}^\top w - \hat{\mu}\|_\infty \le e(n, \alpha_{\log}) + b\}. \tag{2}$$

For oracle rows, $b = 0$. This coordinate relaxation contains the Euclidean confidence region. Weight and fixed-matrix ratio extrema are computed by linear programming.

Target labels and source identities. Reference labels are independently sampled within each known source through a symmetric flip channel with known

$\eta < 1/2$. For m labels per source, define

$$\begin{aligned} h_m &= \sqrt{\log(4/\alpha_{\text{lab}})/(2m)}, \\ L_k &= \text{clip}((\bar{y}_k - h_m - \eta)/(1-2\eta)), \\ U_k &= \text{clip}((\bar{y}_k + h_m - \eta)/(1-2\eta)), \end{aligned} \tag{3}$$

where clipping maps to $[0,1]$. The two rates lie in their intervals simultaneously with probability at least $1-\alpha_{\text{lab}}$. As η approaches $1/2$, the intervals can become vacuous; this is loss of information about θ, not an estimate of aleatoric uncertainty.

From $a > 0$ iid deployment source identities, intersect $\mathcal{W}$ with

$$\begin{aligned} |w_k - \hat{p}_k| \le h_a, \qquad K = 3, \\ h_a = \sqrt{\log(2K/\alpha_{\text{id}})/(2a)}. \end{aligned} \tag{4}$$

At $a = 0$ this restriction is absent. Identities are assumed reliable, including the candidate source. Unlike stratified reference labels, they constrain deployment proportions directly.

Three distinct widths. For a function T on a feasible set S, write $\text{wid}_S(T) = \sup_S T - \inf_S T$. At a positive-mass probe c, define

$$\begin{aligned} T_{\text{cls}}(c; w, A) &= \frac{w_0 A_{0c}}{w_0 A_{0c} + w_1 A_{1c}}, \\ T_{\text{unk}}(c; w, A) &= \frac{w_q A_{qc}}{\sum_k w_k A_{kc}}, \\ \pi_k(c; A) &= \frac{\rho_k A_{kc}}{\rho_0 A_{0c} + \rho_1 A_{1c}}, \\ T_{\text{label}}(c; A, \theta) &= \sum_{k=0,1} \pi_k(c; A)\theta_k, \end{aligned} \tag{5}$$

with fixed reference source weights $\rho = (1/2, 1/2)$. For exact A, W^{cls} and W^{unk} are widths over the feasible deployment weights; W_{label} varies $\theta_k \in [L_k, U_k]$ at the reference anchor. It does not vary deployment weights, so its invariance to n and a is a definition, not independent evidence of disentanglement. These posterior membership widths differ from marginal intervals for w_k.

For estimated A, known cell probabilities vary over boxes $[\max(0, \hat{A}_{kc} - b), \min(1, \hat{A}_{kc} + b)]$. The implementation takes extrema over box corners and projected weight constraints, dropping row-normalization and log/reference coupling. These are conservative outer envelopes, not sharp intervals over the original joint evidence set. Reference inputs can therefore affect all three widths, including the induced anchor $\pi(c; A)$. A feasible zero denominator invokes a vacuous $[0,1]$ policy. The widths are distinct, nonadditive targets.

Validity. Split $\alpha = .05$ equally across log, reference, label, and identity events; unused allocations remain unused. On their intersection, the generating targets lie in the corresponding outer intervals. This is a fixed-budget statement, not a simultaneous guarantee over the swept grid. **[Mateus: Tgus us .... not a ...., I think we should avoid those type of structure, a reviewer could point that sounds a little bit AI.]** Empty sets, coverage, vacuity, and numerical failures are reported separately.

3 IDENTIFICATION LIMITS

With exact source laws and perfect **[Mateus: maybe we should avoid use perfect here]** input information, the identified weight set is $\mathcal{F}(\mu) = \{w \in \Delta_3 : A^\top w = \mu\}$. A target is identified precisely when it is constant on this set. Thus nonidentified weights do not imply positive uncertainty for every functional or cell; the relevant limit is the target's range on $\mathcal{F}(\mu)$.

For the confounded construction, $A_q = (A_0 + A_1)/2$. Writing $s_0 = w_0 + w_q/2$ and $s_1 = w_1 + w_q/2$, every vector

$$(w_0, w_1, w_q) = (s_0 - t/2, s_1 - t/2, t) \tag{6}$$

with $0 \le t \le 2\min(s_0, s_1)$ generates the same input law. No input sample can distinguish these worlds. In our example the exact novel-weight range is $[0.000, 0.730]$. **[Mateus: how we get this values? is not clear to me]**Source identities add constraints directly on these coordinates and can reduce the range below this input-only limit.

At exact A and zero radius, the coordinate and Euclidean constraints both reduce to $A^\top w = \mu$; the computed floors are sharp for this finite-cell model. Novel-membership floors at the two plotted probes are 0.635 and 0.832, and known-membership floors are 0.141 and 0.605. Their means determine the dotted lines in Figure 3. With distinguishable sources, the exact membership floors at these probes are zero. Finite-radius and estimated-reference widths can remain above the floors because of statistical uncertainty and relaxation slack.

3.1 What perfect **[Mateus: maybe avoid perfect here]**logging identifies

The equivalence set separates information about the observable law from information about its latent explanation. In Eq. (6), perfect logging identifies s_0 and s_1 when the two known rows are linearly independent. It does not identify t. More inputs can reduce sampling error around that equivalence set, but cannot select one of its members. This is stronger than an em-

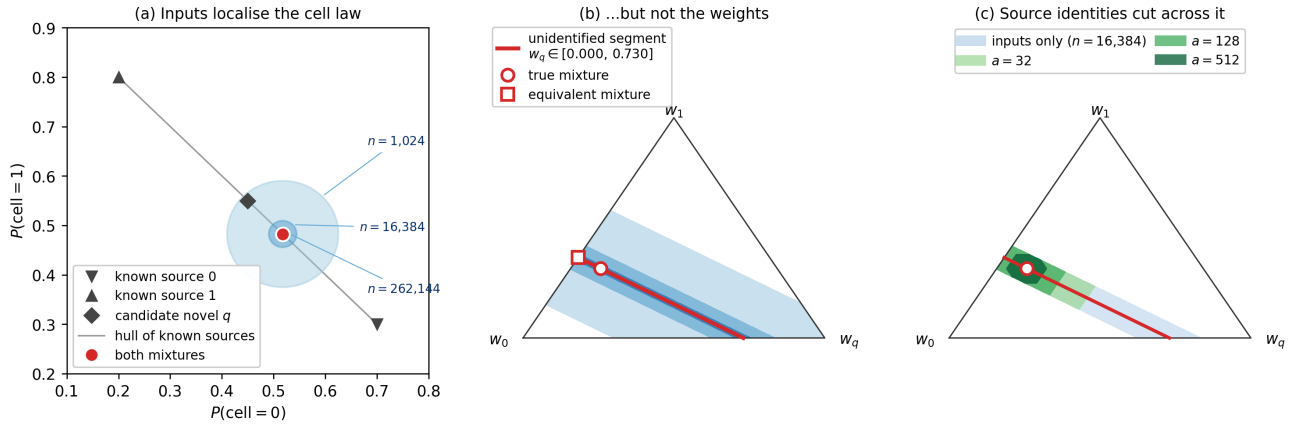


Figure 2: Exact identification geometry. (a) Different mixtures have the same population cell law. (b) Input confidence sets approach a segment, not a point; its exact novel-weight range is $[0.000, 0.730]$. (c) Source-identity bounds intersect that segment directly. The example is evaluated at the population law, with no sampled seeds. Its weight floor differs from cell-membership floors. Reliable identities are assumed; acquisition counts imply no common cost.

pirical plateau: every possible input dataset has the same probability under the equivalent worlds, so an input-only estimator cannot distinguish them even in principle.

The implication is target-specific. Along the confounded segment, the novel posterior is

$$T_{\mathrm{unk}}(c;t) = tA_{qc}/\mu_c. \tag{7}$$

For $\mu_c > 0$, its input-only width is therefore $2\min(s_0, s_1)A_{qc}/\mu_c$. A candidate with no mass at that cell gives zero width even when its marginal weight is unidentified. Known-source membership instead changes through the ratio of $s_0 - t/2$ to $s_1 - t/2$, weighted by the two cell probabilities. It can be constant in special cases; for example, equal effective shares keep the two known weights equal throughout the segment. Neither weight non-identification nor novelty alone guarantees positive width for every target. The positive floors reported here are computed for the selected probes, not assigned universally. **[Mateus: same point of the structure the ..., not ...]**

The distinction also matters when evaluating finite-data intervals. A width above its exact floor may contain sampling error, reference error, and slack from the coordinate relaxation **[Mateus: maybe im being very strict, but also using 3 examples could be pointed as AI]**. Its distance from the floor is not itself a fourth epistemic component. The floor is a population identification statement; a finite-radius width is a conservative summary of the evidence available at that budget. Comparing the two diagnoses a limitation without asserting an additive breakdown.

3.2 Separation controls the value of input evidence

Non-identification and weak separation are different obstructions. For exact source rows, define an affine separation constant on the simplex tangent space:

$$\sigma_A = \inf_{\substack{\|v\|_2=1 \\ \mathbf{1}^\top v=0}} \|A^\top v\|_2. \tag{8}$$

It is positive when the source rows are affinely independent and zero when a nonzero mixture perturbation is invisible in the cell law. In the midpoint construction, $v = (-1/2, -1/2, 1)$ is such an invisible direction after normalization. A confidence radius that vanishes with logging cannot remove that direction.

If $\sigma_A > 0$, two weights in a coordinate set of radius R satisfy $\|A^\top(w - w')\|_\infty \le 2R$. Consequently,

$$\mathrm{diam}_2(\mathcal{W}) \le \frac{2\sqrt{d}\,R}{\sigma_A}. \tag{9}$$

This follows because $w - w'$ has zero coordinate sum and its transformed Euclidean norm is at most $2\sqrt{d}R$. For exact rows, $R = e(n, \alpha_{\log})$, so the bound shrinks with the input radius. Combined with positive probe-denominator bounds, it controls membership widths through the same ratio argument used for source identities. The constant can be large when the sources are close; identification in the population does not imply useful intervals at modest sample sizes.

For estimated rows, the projected weight constraints use $\hat{A}$ and $R = e(n, \alpha_{\log}) + b$. The same geometric inequality applies to that supplied matrix when its separation constant is positive, but b does not vanish when

only n grows. In addition, the posterior envelope varies the rows themselves. Thus an input sweep with fixed reference evidence can stop improving before reaching the exact-row target floor. A sampled reference matrix can also have different conditioning from its population counterpart. This is why oracle-reference and exact-law controls answer different questions: one removes finite reference error, while the other removes both reference and deployment sampling error.

These inequalities describe the finite-cell construction, not a universal relation between image similarity and epistemic uncertainty **[Mateus: also the structure These ...., not a...]**. Separation is measured after the specified partition; two visually distinct classes can be indistinguishable in an uninformative representation. The selected weak-separation control has positive population separation but insufficient reference precision on the swept budgets. The exact confounded construction instead has a genuine zero separation direction. A visually flat curve alone cannot distinguish those explanations.

3.3 Why the acquisition matters

A source-stratified reference label reveals information about θ_k, conditional on a source whose identity is already known. Its sampling proportions are chosen by the reference design and are not observations of deployment prevalence. Under this model, changing m therefore cannot restrict t in Eq. (6). Conversely, an unlabelled deployment cell has a law independent of θ, so it cannot replace reference target labels. These are statements about the specified observation schemes, not about all possible uses of labels. Deployment target labels with an informative source-dependent target law would constitute another acquisition and could constrain weights; we do not evaluate that scheme here.

Reliable deployment identities reveal w directly. They distinguish the equivalent worlds even when their input laws agree. For exact A, suppose the known and total probe denominators are bounded below throughout the feasible set by $\gamma_{\mathrm{cls}} > 0$ and $\gamma_{\mathrm{unk}} > 0$. The identity box gives $\|w - w'\|_\infty \le 2h_a$ for any two retained weights. The elementary ratio bound $|x/s - x'/s'| \le (|x - x'| + |s - s'|)/\gamma$ for $0 \le x \le s$ then yields

$$W_c^{\mathrm{cls}} \le \min\{1, 2h_a(2A_{0c} + A_{1c})/\gamma_{\mathrm{cls}}\},$$
$$W_c^{\mathrm{unk}} \le \min\{1, 2h_a(A_{qc} + \sum_k A_{kc})/\gamma_{\mathrm{unk}}\}. \quad (10)$$

These bounds show how direct identity information can narrow membership targets without input-law identifiability. The denominator conditions are essential: low-mass cells can amplify weight error, and an admissible zero denominator triggers the stated vacuous policy. With uncertain source rows, identity observations alone need not remove the remaining reference uncertainty.

For exact A, the reference-anchor label width is $\sum_{k=0,1} \pi_k(c; A)(U_k - L_k)$, bounded by $\min\{1, 2h_m/(1 - 2\eta)\}$. This explains its response to reference labels and its lack of dependence on deployment weights. These exact-row bounds are consequences of the chosen confidence constraints; they do not supply sharp rates for the estimated-reference envelopes used in the main experiment.

4 ACQUISITION EXPERIMENTS

Protocol. The frozen protocol compares distinguishable, weakly separated, and hull-confounded categorical worlds. Each seed generates independent population streams for deployment cells, known-source reference inputs, noisy target labels, and deployment identities. Budgets use nested prefixes within each stream. The main figure fixes $r = 32{,}768$ inputs per known source and baseline $(n, m, a) = (4096, 512, 0)$, with $\eta = .2$, then varies one acquisition at a time. The candidate row is exact. Two positive-support probes are averaged within each of 64 independent seeds; pointwise bootstrap intervals describe the estimated mean width, not the epistemic interval itself. The protocol is an exploratory follow-up, not a historical preregistration. **[Mateus: last 2 phrases uses the structure of something ..., not something... maybe we should avoid it]**

Distinguishable sources. Increasing n from 256 to 65,536 reduces known-membership width from .913 to .178 and novel-membership width from .222 to .027. Increasing target labels from 32 to 8192 per source reduces W_{label} from .709 to .091, while the two membership widths stay fixed. Deployment identities reduce known-membership width from .347 at $a = 0$ to .077 at $a = 8192$. These responses reflect the information supplied by the specified acquisitions.

Confounded sources. The same input increase narrows known-membership width from 1.000 to .454, but novel membership only from 1.000 to .886. Reference target labels leave both membership widths at 1.000 under the baseline deployment evidence. Just 32 deployment identities reduce novel-membership width to .483; 8192 reduce it to .049. The latter channel crosses the exact input-only floor because it observes source composition. This contrasts information types, not acquisition efficiency at equal cost. **[Mateus: same feedback on this ..., not ...]**

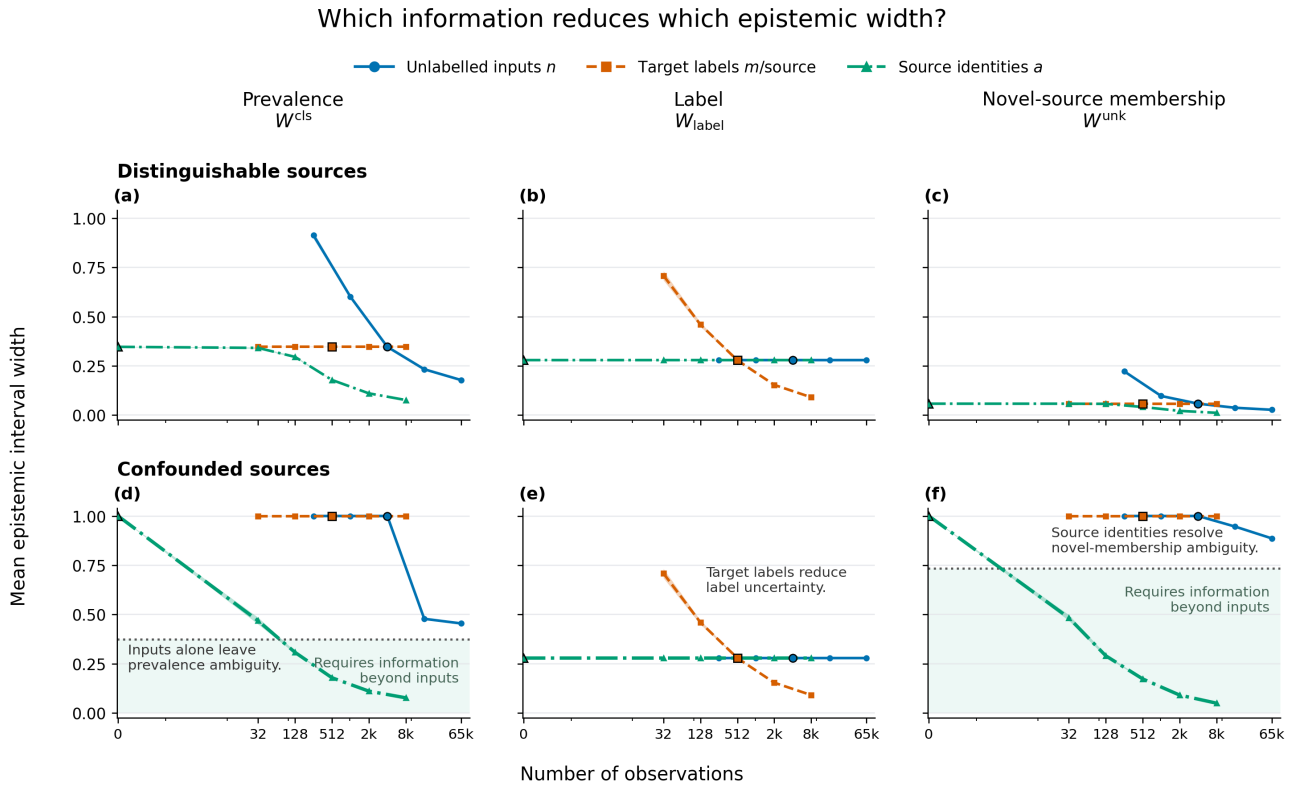


Figure 3: Acquisition-dependent reducibility. Columns are distinct epistemic targets; rows contrast distinguishable and confounded sources. Curves vary n, reference labels m/source, or deployment identities a, holding other budgets at the common baseline. Black outlines at $n = 4096$, $m = 512$, and $a = 0$ mark that same configuration at different horizontal positions. Bands are pointwise 95% bootstrap intervals for mean widths across seeds. Dotted lines are exact input-only floors under perfect reference information; shading marks widths requiring information beyond inputs. Counts are not equal costs **[Mateus: this statement is relevant?]**. Reliable candidate-source annotations are assumed.

Cross-effects. Figure 4 varies known-source reference inputs while holding deployment and target-label evidence fixed. Under distinguishable sources, all three widths fall; W_{label} changes because its cell-dependent anchor depends on A. Joint reference input–label acquisition reduces it further. Under confounding, membership widths remain broad at the fixed $n = 4096$, reaching .935 and .998. These are finite-budget endpoints, not the exact floors, and retain both reference and deployment uncertainty. The relationship between acquisitions and widths is many-to-many.

Coverage and controls. In every displayed configuration, all 64 seeds retain the generating targets on the joint confidence event; no empty sets or numerical failures occur. An observed 64/64 rate has one-sided 95% lower bound .954 at a fixed configuration, not a guarantee of perfect coverage **[Mateus: this is .... not ..., and the argument about perfect coverage, what we mean by perfect here? maybe we should remove thsi statement]**. Exact zero-radius controls distinguish structural floors from finite-evidence plateaus. In the weakly separated world, estimated-reference known-membership widths remain vacuous across the input sweep, while oracle references yield contraction; Section D reports the contrast.

4.1 Reading the comparison

The horizontal axis records observations, not a common resource **[Mateus: again the ..., not a ...]**. A source annotation may be cheaper when provenance metadata already exist and more expensive when it requires investigation; a reference target label may have a different collection cost again. The curves identify which information constrains an ambiguity, rather than an optimal purchase under a specified budget **[Mateus: even we may very to the counterpoint using rather than, i think we should avoid it]**. The black-outlined points share the same evidence despite appearing at three horizontal positions. Comparing equal x coordinates would compare different exper-

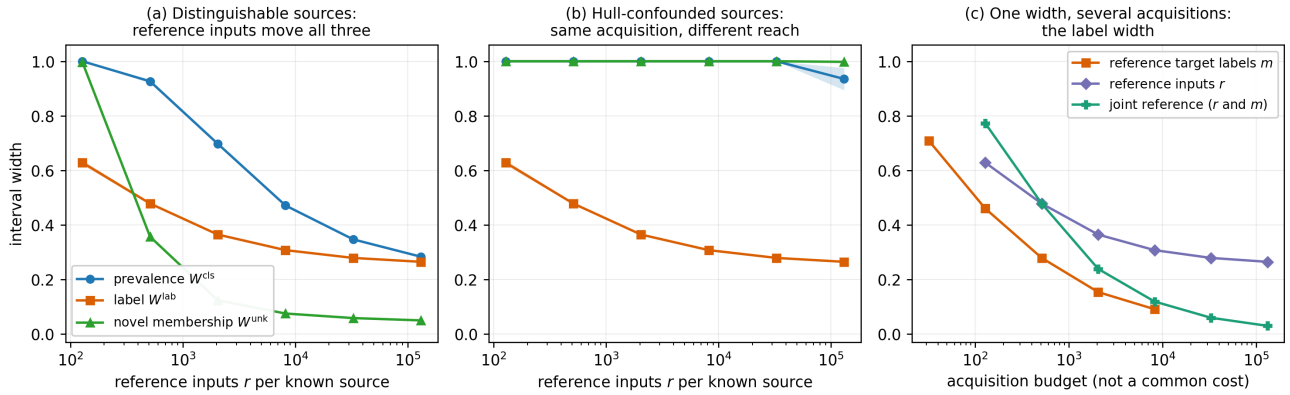


Figure 4: Reference evidence creates cross-effects. (a) Reference inputs alone reduce all three widths with distinguishable sources. (b) Under confounding, the label width falls but membership widths remain broad at the fixed deployment-log budget; these endpoints are not identification floors. (c) Target labels, reference inputs, and joint reference pairs each affect W_{label}. Bands describe mean-width uncertainty across seeds; horizontal counts imply no common cost.

imental configurations.

The fixed reference anchor also determines what the middle column can establish. Its flat response to logging is expected from the definition, whereas the lower-row input-only floors concern genuine observational equivalence. Figure 4 prevents interpreting the three columns as independent channels: reducing uncertainty in A changes the anchor as well as the weight evidence. A label-width change from reference inputs is consequently compatible with label counts remaining fixed. It is neither an estimate of aleatoric uncertainty nor evidence that the target labels were used to estimate input conditionals.

Finally, width and validity answer different questions. Coverage checks whether the interval retains the specified target; informativeness checks whether it rules anything out. A vacuous interval can have perfect coverage. Conversely, a narrow point interval under confounding can fit the observed input law while excluding a generating weight. We retain both diagnostics and compare finite-data results against exact controls. The weakly separated reference control is especially useful here: its positive finite-budget widths occur despite a zero exact floor, showing why visual flatness alone cannot establish an identification limit.

What the average width summarizes. The figures average the same two probe cells with equal weights. They do not weight probes by deployment prevalence, and should not be read as a population risk or a calibrated classifier confidence score **[Mateus: again the counterpoint. It seems that we are justifying too much. it is sounding not confident of what we are proposing]**. Cell-level outputs are retained because the identification floors differ across probes: the novel-membership floor is .635 at one and .832 at the other. A mean can summarize the acquisition contrast while concealing a cell that remains poorly resolved. The deployment sampling law and the plotting weights therefore serve different purposes.

Within a seed, repeated budgets reuse stream prefixes, so the curves are paired rather than independent observations. The seed is the replication unit for uncertainty bands and endpoint comparisons. The confidence allocations apply at a specified budget; inspecting the full curve and selecting a budget afterward does not inherit a uniform guarantee. Our reported zero failure rates describe the saved configurations. They do not justify stopping at the first apparently narrow interval or certify every possible input cell. Such an acquisition policy would require its own sampling and error-control argument.

4.2 Paired endpoint contrasts

Table 1 reports the change between the first and last budgets of each selected acquisition. The same seed supplies both endpoints, so each contrast is formed within seed before averaging; cells are not treated as independent replicates. The bootstrap intervals quantify uncertainty in this mean contrast across seeds. They are not confidence intervals for an individual input's posterior, and are not simultaneous across rows.

The contrast in novel membership is especially direct. In the confounded world, the full input sweep leaves width .886, whereas deployment identities reduce it to .049. The paired changes measure the responses over

World	Acquisition	Target	Budgets	Change	95% paired interval
Distinguishable	Inputs	W^{cls}	256–65,536	-0.736	$[-0.753, -0.718]$
Distinguishable	Inputs	W^{unk}	256–65,536	-0.195	$[-0.201, -0.190]$
Distinguishable	Reference labels	W_{label}	32–8,192	-0.618	$[-0.641, -0.595]$
Distinguishable	Source identities	W^{unk}	0–8,192	-0.046	$[-0.047, -0.046]$
Confounded	Inputs	W^{cls}	256–65,536	-0.546	$[-0.547, -0.545]$
Confounded	Inputs	W^{unk}	256–65,536	-0.114	$[-0.116, -0.112]$
Confounded	Reference labels	W_{label}	32–8,192	-0.618	$[-0.641, -0.595]$
Confounded	Source identities	W^{unk}	0–8,192	-0.951	$[-0.951, -0.950]$

Table 1: Paired mean-width changes, last minus first budget, across 64 seeds with estimated known-source rows. Negative changes indicate narrowing. Intervals are read from the archived endpoint bootstrap; unused evidence is held at the common baseline. Acquisition budgets have different units and costs.

each sweep; the identification argument, rather than the confidence interval for a change, explains why inputs cannot eventually substitute for source identities. Reference target labels yield the same label-width response in both worlds because the reference laws and label sampling are unchanged. This shared response is a protocol control, not a second independent demonstration of a label–prevalence decomposition.

4.3 Using the acquisition diagnosis

The results suggest a sequence of questions rather than a universal acquisition rule. First, specify the target: a marginal source proportion, known-source membership at a probe, and a reference-anchored target probability need not have the same identification status. Next ask which worlds remain compatible with the available observations. A broad membership interval with a positive exact input-only floor calls for information that distinguishes those worlds; increasing an input budget alone cannot accomplish that. When the exact floor is zero, a broad computed interval may instead indicate insufficient observations, weak separation, or conservative reference propagation. The oracle-reference control tests one of these explanations in our synthetic setting.

Finally, distinguish an informative annotation scheme from an available one. Source identities are useful in the confounded construction because they are reliable deployment observations. Annotating an equal number of examples from each source by design would describe a reference pool, not the incoming mixture, and would not have the same effect. Likewise, a candidate source that is unspecified cannot be given a calibrated membership interpretation by simply adding an extra slot. The present results assume that candidate's feature law is fixed; they do not provide an acquisition strategy for learning or discovering it.

These restrictions delimit what the figure can recommend. It compares what information does under one controlled observation model. It does not rank annotation policies at equal cost, establish adaptive stopping thresholds, or infer a cause from an empirical plateau alone. A deployment application would additionally need a declared target, a justified sampling design, and a validation of the reference and annotation assumptions. Those requirements are why the synthetic comparison is presented as an identification and information study rather than a clinical or image-classification benchmark.

5 RELATED WORK AND SCOPE

Label-shift quantification estimates deployment prevalences under stable source conditionals. BBSE uses a predictor's confusion matrix and requires invertibility (Lipton et al., 2018). Quantification confidence intervals and extensions incorporating deployment labels are studied by Vaz et al. (2019). Distribution feature matching unifies prevalence estimators and analyzes robustness to unknown contamination (Dussap et al., 2023). REACH's declared candidate is a narrower assumption than arbitrary contamination. Our pseudo-inverse comparator is a finite-cell illustration, not a reproduction of published BBSE or an identification-aware baseline. Appendix comparisons motivate retaining ambiguity; they do not establish superiority over those methods.

The evidence is limited to a fixed partition, two known sources, a specified novel row, a binary target, and reliable source annotations. Conditional independence and known label noise are explicit assumptions. The exact-row width bounds do not characterize the estimated-reference envelopes; those responses are measured on finite grids. The radius and reference envelopes are conservative, and confidence statements are not valid for adaptive budget selection without a further argument. CIFAR images supply an illustra-

tion, not real-data validation. Candidate misspecification is examined only in the qualified stress test of Section C. Aleatoric uncertainty is outside the scope.

6 CONCLUSION

REACH characterizes which epistemic uncertainties a given acquisition can reduce and which ambiguities require another information source. Exact source confounding leaves positive membership widths even with perfect input-law knowledge. Deployment identities address that ambiguity directly; source-stratified target labels address the reference target law. Reference inputs can affect several widths. The useful decision is therefore to specify the uncertainty target and the missing knowledge before choosing what additional information to collect. **[Mateus: maybe is not too short?]**

AI USE STATEMENT

An AI assistant was used for implementation, writing, figure preparation, and auditing. Numerical results were produced by the accompanying code. The authors remain responsible for all claims, assumptions, references, and validation.

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A EVIDENCE BOUNDS AND COMPUTATION

For a categorical sample, $\mathbb{E}\|\hat{\mu} - \mu\|_2 \leq 1/\sqrt{n}$ by the multinomial variance and Jensen's inequality. Replacing one observation changes this norm by at most $2/n$. McDiarmid's bound therefore gives $\Pr\{\|\hat{\mu} - \mu\|_2 > e(n, \beta)\} \leq \beta$. For estimated rows, triangle inequality and nonnegative weights summing to one bound the mixture reference error by the maximum row error b. Since the infinity norm is no larger than the Euclidean norm, Eq. (2) retains the generating weights on the allocated evidence events. Hoeffding's inequality and union bounds give Eqs. (3) and (4). Their intersection has probability at least $1 - \alpha$.

With fixed rows, linear-fractional extrema use the Charnes–Cooper transformation. If a feasible weight vector gives zero probe mass, the conditional target is not determined by the joint law and the implementation returns [0,1]. Estimated-reference envelopes evaluate cell-box corners; they relax joint constraints and need not be sharp. None of these statements ensures nested realized sets as sample size increases: empirical centers change with the observations.

A correctly specified family retains truth on its confidence event. Outside the event, a nonempty set can exclude truth, and an empty set can result from sampling error. Declination is therefore reported separately. If M is an answered miss and D declination, the unconditional bound $\Pr(M) \leq \alpha$ only gives $\Pr(M \mid \neg D) \leq \alpha / \Pr(\neg D)$. The archived summaries include both answered-only coverage and unconditional answered-miss rates.

B SUPPORTING PREVALENCE COMPARISONS

These comparisons concern a marginal known-source weight, not the cell-membership widths of Figure 3. Each budget uses 120 seeds and nominal $\alpha = .05$. The comparators are EM percentile bootstrap and an unconstrained pseudo-inverse with a multinomial Wald interval. The latter returns rank/condition diagnostics but does not act on them to decline. Its use under rank deficiency falls outside the assumptions of invertible-matrix BBSE. The bootstrap checks unsupported observations in the original log before resampling.

B.1 Identified source weights

With separated rows, all methods answer. Credal-to-bootstrap width ratios are 3.67, 3.67, 3.71, and 3.68 over a 64-fold input range. This describes the tested grid, not an asymptotic constant, optimality, or dominance between methods with different guarantees.

B.2 Confounded source weights

The exact known-source weight set is [.270,.635], with width .365. EM selects a point within that set but attaches an interval that contracts around it; at the largest budget its width is .017 and coverage of the generating weight is zero. REACH retains a width .446. The comparison illustrates false precision from an identification-unaware interval, not inconsistency of EM's observable-law fit.

Observed coverage of 120/120 has one-sided 95% lower bound .975; zero of 120 has upper bound .025. Neither empirical endpoint is an absolute probability. Different budgets reuse seed indices and are not pooled as independent repetitions.

B.3 Unsupported source family

The log includes cells unsupported by a known-only family. EM declines by inspecting the original sample. The coordinate polytope becomes empty at the larger budgets; a pseudo-inverse continues answering even when its weights leave the simplex. This detects support misspecification, which is different from the equivalent-world ambiguity in the main text.

Euclidean residual flags and coordinate-polytope emptiness are different tests: a residual .098 against radius .090 can flag a Euclidean incompatibility while the coordinate polytope remains nonempty. They should not be reported as interchangeable detections.

C CONDITIONAL FINITE-POOL STRESS TEST

This constructed-deployment experiment uses Gaussian features passed through the finite-reference pipeline. It uses 32 replicates, 16,384 reference rows per known source, and 131,072 deployment inputs.

Zero novel mass is excluded at .25 but not at .05 or .10 on the tested grid. This is not an estimate of a universal detection threshold. The radius .0427 comprises reference error .0320 and log error .0107; the displayed configuration does not establish general minimum data requirements. Coverage includes vacuous intervals; informativeness is reported separately. In the confounded arm at novel mass .25, coverage is 1 while informative rate is 0.

Two qualifications prevent a population-coverage interpretation. The target/cell independence gate has adjusted $p = .016$ for one known source and fails in the saved summary; a false rejection is possible, and

Table 2: Identified weights: empirical coverage and mean interval width; no declines.

	Credal		EM bootstrap		Unconstrained	
n	coverage	width	coverage	width	coverage	width
1,024	1.000	0.431	0.950	0.117	0.958	0.120
4,096	1.000	0.215	0.950	0.059	0.967	0.060
16,384	1.000	0.108	0.925	0.029	0.942	0.030
65,536	1.000	0.054	0.933	0.015	0.958	0.015

Table 3: Confounded weights: empirical coverage, mean width, and bootstrap midpoint for true known-source weight .585.

	Credal		EM bootstrap			Unconstrained	
n	coverage	width	coverage	width	midpoint	coverage	width
1,024	1.000	0.855	0.200	0.135	0.484	0.033	0.122
4,096	1.000	0.686	**0.000**	0.068	0.479	**0.000**	0.061
16,384	1.000	0.527	**0.000**	0.034	0.477	**0.000**	0.031
65,536	1.000	0.446	**0.000**	0.017	0.476	**0.000**	0.015

rerunning seeds until acceptance would not repair it. The candidate row is estimated from a finite evaluation pool with no separate error term. Replicates share generated pools and a partition, and source counts are fixed by rounding rather than iid mixture draws. Results are conditional on this construction. The main acquisition experiment instead samples independent population streams and fixes the candidate row exactly.

Using the misspecified uniform candidate yields broad or vacuous intervals at lower doses and infeasibility at .25, with residual .112 exceeding radius .0427. A useful novel-membership interval therefore depends on an adequately specified candidate; automatic candidate discovery is not provided by REACH.

D ADDITIONAL ACQUISITION CONTROLS

Weak separation. With estimated reference rows, known-membership widths stay at 1.000–1.000 across the input sweep, with vacuity rate 1.00. Oracle rows instead yield 1.000–0.774. Deployment identities give estimated-reference width 0.085 at the largest budget. The exact floor is zero: the finite-budget obstruction is weak separation and reference uncertainty, not exact confounding.

Equivalent target laws. Two worlds with identical source laws and weights but different source-specific target rates have identical unlabelled input laws. With target rates unconstrained, the auxiliary deployment-target interval remains [0,1]. Reference labels constrain those rates, but interval separation depends on the probe gap and residual weight uncertainty. For the larger gap (.431), all 32 replicates separate at $m \geq 128$ with the largest input budget, versus $m \geq 512$ at $n = 1024$. The smaller gap (.105) never separates on the swept grid. This auxiliary target varies deployment weights and a free candidate target rate; it differs from the fixed-anchor W_{label}. A second pair keeps target rates fixed and changes novel mass through the confounded construction; reference target labels cannot identify that mass.

Noise sensitivity. Equation (3) describes analytic sensitivity to a known flip rate: the raw uncertainty is amplified by $1/(1-2\eta)$ before clipping. The main acquisition sweep holds $\eta = .2$ fixed. This equation is not presented as an additional varying-noise experiment or an AU estimate.

E REPRODUCTION AND DATA

The protocol is frozen in `protocols/acquisition_v1.json`; each bundle stores its settings, per-seed outputs, and source hashes. Figure scripts read those bundles without rerunning training. From the project root:

```
python main.py identifiability --seeds 120
python main.py realsim --repeats 32
python main.py equivalence --seeds 32
python -m train.acquisition_curves
python paper/plot_main_figure.py
python paper/plot_identification_geometry.py
python paper/plot_width_distinctness.py
```

Table 4: Unsupported family: decline rate and empirical coverage conditional on answering.

	Credal		EM bootstrap		Unconstrained	
n	declines	coverage	declines	coverage	declines	coverage
1,024	0.000	0.983	1.000	—	0.000	0.925
4,096	1.000	—	1.000	—	0.000	0.850
16,384	1.000	—	1.000	—	0.000	0.625
65,536	1.000	—	1.000	—	0.000	**0.133**

Table 5: Conditional finite-pool stress test: mean novel-mass interval, zero exclusion, and target-rate coverage at the displayed budgets.

dose δ	mean interval for novel mass	excludes zero	target-rate coverage
0.00	[0.000, 0.146]	0.00	1.00
0.05	[0.000, 0.196]	0.00	1.00
0.10	[0.000, 0.246]	0.00	1.00
0.25	[0.112, 0.396]	1.00	1.00

```
python paper/make_numbers.py --check
python main.py test
```

The regression suite contains 33 tests. From `paper/`, compile with `tectonic main.tex -keep-logs`, adding `-keep-intermediates` to retain the page-boundary labels. Then run `python check_page_limits.py` from the same directory. The TikZ illustration is included from its source and five CIFAR-10 test images (indices 8, 46, 16, 39, 11); no CIFAR performance is estimated.

All experimental data are synthetic. Independent streams, nested budgets, allocations, probe choices, and mean-width bootstrap settings are documented in the frozen protocol. Per-configuration summaries retain confidence-event inclusion, target coverage, vacuity, empty-set, and numerical failure rates; mean-width bands do not replace those diagnostics. No uniform guarantee over budgets or cells beyond the stated allocations is asserted.